

Philippines Food Price Analytics

Findings & Analysis Report
Python / pandas • SQL • Power BI

Project Type	End-to-end Data Analytics Project
Dataset	Philippine food price data based on the WFP Price Database
Primary Tools	Python, pandas, MySQL/SQL, Power BI, DAX
Purpose	Analyze food-price trends, regional differences, retail premiums, and price movement

Prepared as a portfolio project to demonstrate practical data preparation, business analysis, data visualization, and analytical communication.

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1. Executive Summary

This project analyzes Philippine food-price data through an end-to-end analytics workflow. Python and pandas were used to automate extraction, cleaning, transformation, and data-quality processing. SQL was then used to answer business-oriented analytical questions, while Power BI was used to communicate the results through an interactive dashboard.

The analysis focuses on how food prices change over time, how prices differ between wholesale and retail markets, how prices vary by region, and which commodities show larger price movements or greater dispersion. The project is designed to demonstrate not only technical tool usage, but also the ability to translate raw data into questions that support business decision-making.

2. Business Questions

- How have commodity prices changed from one year to another?
- Which commodities experienced price increases?
- What is the observed price range for each commodity?
- How large is the difference between wholesale and retail prices?
- Which regions have prices above or below the Philippine benchmark?
- Which commodities show greater price volatility?
- How does the latest recorded price compare with the previous recorded price?

3. Data Source and Dataset

The project uses a Philippine food-price dataset derived from the World Food Programme (WFP) Price Database and distributed through Kaggle. The dataset contains observations covering food commodities and market prices across Philippine administrative areas and markets.

Column	Role in analysis
date	Observation date / time-series analysis
admin1	Region-level analysis
admin2	Province/city-level detail
market	Market-level context
category	Commodity category
commodity	Food item being analyzed
unit	Measurement unit
pricetype	Wholesale or Retail
price	Recorded local price
usdprice	USD price representation

4. Methodology

4.1 Python / pandas ETL

The Python component automates the preparation of the raw CSV data. The workflow was developed to identify input files, extract the dataset, perform cleaning and transformation operations, and prepare a reliable dataset for downstream SQL and Power BI analysis.

ETL flow: Extract → Transform → Validate → Load/Export

4.2 SQL Business Analysis

SQL was used to convert the cleaned data into business-oriented metrics. The analysis uses CTEs, conditional aggregation, aggregate functions, date conversion, and window functions such as LAG() to compare observations across time.

A key analytical principle used throughout the project was to aggregate observations to the appropriate business grain before comparing them. For example, the latest-versus-previous price analysis first calculates a daily average by commodity and then applies a chronological LAG(). This avoids comparing arbitrary records from different regions or markets.

4.3 Power BI

Power BI was used to create an interactive dashboard with a dedicated DateTable, DAX measures, KPI cards, trend analysis, regional comparisons, commodity rankings, and wholesale-versus-retail analysis. The date table enables year-based filtering and year-over-year calculations.

5. Key Findings

The following findings are based on the analytical work completed during the project and the dashboard states discussed during development. Exact values can change when different commodities, regions, price types, units, or date filters are selected.

Overall price indicators

A dashboard snapshot during development showed an Average Price of 113.83 and a Latest Price of 140.45. These values illustrate that the latest observed period can differ materially from the overall average depending on the active filter context.

Year-over-year movement

The dashboard's YoY calculation was validated using year selection. For Sweet Potato leaves, the 2022 comparison showed approximately +17.09% growth relative to 2021. A 2023 comparison was discussed at approximately +10.5% relative to 2022.

Wholesale versus retail

The analysis demonstrates that retail prices can be higher than wholesale prices. The Retail Premium % metric measures the relative gap using wholesale price as the baseline, allowing commodities to be compared on a percentage rather than only an absolute price difference.

Regional differences

Regional analysis compares each region's average price with a Philippine benchmark. This makes it possible to identify regions that are relatively higher or lower than the benchmark and quantify the difference.

Price volatility

Standard deviation was used as a measure of price dispersion. A higher standard deviation means prices are more spread out around the average. The analysis also established that volatility should not be labeled Stable/Unstable by simply comparing standard deviation with the average price; the two statistics describe different concepts.

Latest versus previous price

The latest-price analysis required first aggregating multiple observations per commodity and date. After daily averages were created, LAG() was applied chronologically to obtain the previous recorded daily average. This produces a more meaningful business comparison than applying LAG() directly to raw multi-location observations.

Commodity coverage

The dashboard also showed that individual commodities do not necessarily cover the full historical period. For example, Rice (Special) was observed in the dashboard over 2020–2023. Missing years for a commodity therefore should not automatically be interpreted as zero prices or a price decline.

6. Power BI Dashboard Design

KPI	Purpose
Average Price	Shows the average price under the current filter context.
Latest Price	Shows the average price for the latest available date in context.
YoY Growth %	Measures the percentage change relative to the previous year.
Retail Premium %	Measures how much higher/lower retail price is relative to wholesale.

Primary visuals: Price Trend Over Time; Average Price by Region; Commodity Price Ranking; Wholesale vs Retail comparison with Retail Premium %. The dashboard is intended to answer three questions quickly: How are prices moving? Where are prices relatively high or low? Which commodities have the largest wholesale-to-retail gaps?

7. Business Implications

- Price monitoring should consider both the absolute price and the percentage change because commodities operate at different price levels.
- Regional benchmarking can help identify areas where food prices are relatively higher and may require additional investigation.
- Wholesale-to-retail gaps can reveal where consumer prices diverge from upstream market prices.
- Volatility is useful for identifying commodities with less predictable prices, which may matter for procurement and budgeting.
- Latest-versus-previous comparisons provide a short-term signal that can complement longer-term YoY trends.

8. Recommendations

- Monitor commodities with repeated increases rather than reacting to a single-period movement.
- Investigate regions with consistently high prices relative to the national benchmark.
- Track retail premium percentages alongside wholesale prices to understand consumer-facing price pressure.
- Use volatility or coefficient of variation to identify commodities that require closer monitoring.
- When interpreting missing years, distinguish between 'no recorded observation' and 'zero price'.

- For future development, add automated data-quality summaries and scheduled ETL execution to make the pipeline more operational.

9. Limitations

- The dataset contains observations from different regions, markets, dates, units, and price types. Results depend on the selected analytical grain and filters.
- A national average can be calculated in different ways. Averaging regional averages gives each region equal weight, while averaging raw observations weights records according to their number of observations.
- Observed minimum and maximum prices can come from different dates or locations and therefore represent an observed range, not necessarily a market range at one point in time.
- The dashboard is descriptive and does not establish causal relationships between price changes and external factors such as inflation, weather, supply disruptions, or policy changes.

10. Conclusion

This project demonstrates an end-to-end approach to data analytics using a real-world Philippine food-price dataset. The Python component automates data preparation, SQL converts the data into business-focused analyses, and Power BI presents the results in an interactive format.

The most important outcome is not a single price metric, but the analytical workflow: defining a business question, selecting the appropriate level of aggregation, calculating a defensible metric, validating the result, and communicating the finding clearly. This workflow reflects practical skills required in a Data Analyst role.

Data source: Philippines Food Prices Dataset, based on the World Food Programme (WFP) Price Database, accessed through Kaggle. This report summarizes the analytical workflow and findings developed for the portfolio project.